Practical 2: Working with Users,Groups, and Permissions.

- **1. Users:-**

```
sudo adduser user1          # Creates a new user
sudo passwd user1           # Sets or changes password
sudo passwd -l user1        # Locks the user account
sudo passwd -u user1        # Unlocks the user account
sudo deluser user2          # Deletes the user
su user1          # Switches to user1
whoami            # Displays current logged-in user
id user1          # Shows user ID and group info
who           # List logged-in users
```

- **2. Groups:-**

```
sudo addgroup group1            # Creates a new group
sudo usermod -aG group1 user1            # Adds user1 to group1
sudo usermod -aG group1,group2 user1          # Add multiple groups
groups user1                    # Shows groups of user1
sudo gpasswd -d user1 group1   # Removes user1 from group1
cat /etc/group         # List all groups
sudo delgroup group1           # Delete group   OR
sudo groupdel group2
grep group1 /etc/group         # View group details
```

🔹 3. Permissions:-

```
mkdir exam                    # Creates a directory named ‘exam’
touch exam/file1.txt          # Creates a file
ls -l                         # View file permissions
```

Syntax sequence of three digits: user.owner group all

```
chmod 755 exam                # Numeric permission (rwx r-x r-x)
chmod 644 exam/file1.txt      # Numeric permission (rw- r– r--)

chmod u+x exam/file1.txt      # Add execute permission to user
chmod g+w exam/file1.txt      # Add write permission to group
chmod o-r exam/file1.txt      # Remove read permission from others

sudo chown user1 exam         # Change owner to user1
sudo chown user1:group1 exam  # Change owner and group

chmod -R 755 exam             # Apply permissions recursively

umask                         # View default permissions
umask 022                     # Set default permissions
```

Practical 3:

Initial settings: Add a User, Network Settings, Change to static IP address, Disable IPv6 if not needed, Configure Services, display the list of services which are running, Stop and turn OFF auto-start setting for a service if you don't need it, Sudo Settings.

1. Add User

sudo adduser user1
id user1

2. Check Network

ip a **OR** ifconfig

3. Set Static IP

sudo nano /etc/netplan/01-netcfg.yaml

Example:

```
network:
 version: 2
 ethernets:
   ens33:
     dhcp4: no
     addresses: [192.168.1.100/24]
     gateway4: 192.168.1.1
     nameservers:
       addresses: [8.8.8.8]
```

Apply:

sudo netplan apply

4. Disable IPv6 (OPTIONAL)

sudo nano /etc/sysctl.conf

Add:

net.ipv6.conf.all.disable_ipv6 = 1

5. Install & Start Service (SSH)

sudo apt update
sudo apt install openssh-server
sudo systemctl start ssh
sudo systemctl enable ssh

6. Show Running Services

systemctl list-units --type=service --state=running (shows only running services)
OR
service --status-all (Shows all services (running + stopped))

7. Stop & Disable Service

sudo systemctl stop ssh
sudo systemctl disable ssh

8. Sudo Access

sudo usermod -aG sudo user1
groups user1

Practical 4: SSH

sudo su
sudo apt-get update
sudo apt-get install openssh-server
sudo apt install net-tools
Ifconfig
sudo systemctl start ssh
sudo systemctl enable ssh
sudo systemctl status ssh

Run as administrator the Windows CMD on your real PC.
ping [IP address of virtual machine]

Install Putty, and connect to the virtual machine by entering its IP address as the Host Name.

*ls
sudo su
tcpdump -e

If error occurs only then do this: Alternative for ssh install :

You tried:

sudo systemctl start sshd

But in most Ubuntu systems:

- Service name is **ssh**, not sshd

1. Install SSH server

sudo apt update

sudo systemctl start ssh

sudo systemctl enable ssh

sudo systemctl status ssh

You should see:

active (running)

Verify your IP (you already did)

From your ifconfig:

inet 192.168.232.128

Test SSH (from same machine)

ssh system1@192.168.232.128

Check the real status

Run:

sudo systemctl status ssh.socket

If you see:

Active: active (listening)

Then SSH is **working perfectly**

When ssh.socket is also inactive

sudo systemctl disable ssh.socket

sudo systemctl stop ssh.socket

sudo systemctl enable ssh

sudo systemctl start ssh

Now: systemctl status ssh

Practical 5: FTP

Sudo su
Sudo apt-get update
Sudo apt-get install vsftpd
Sudo nano /etc/vsftpd.conf
Make the following changes:

1) Listen = YES
2) Listen_ipv6 = NO
3) Anonymous_enable = YES
Uncomment
4) write_enable = YES
Uncomment
5) chroot_local_user = YES (MAKE SURE YOU ARE UNCOMMENTING THE SECOND ONE; HUGE ERRORS MAY OCCUR IF NOT)

6)allow_writeable_chroot=YES (write this)

Now, review the config file with
sudo cat /etc/vsftpd.conf

Sudo systemctl restart vsftpd
Sudo systemctl status vsftpd
ifconfig

Now, login with:-

ftp siddhant@192.168.136.137 (i.e. [username]@[ip address])
It will prompt you for the password. Enter it to login.

Open file explorer in Windows:
To access ftp site in the address bar type ftp url:
ftp://username:password@inet
ftp://system:system123@192.169.136.131
ftp://username@ip and then when prompted enter pswrd.

Practical 6: DNS and BIND

Sudo apt install bind9
sudo apt install dnsutils
Confirm this one with y!

Sudo nano /etc/bind/named.conf.options

UNCOMMENT the “forwarders” dictionary (REMOVE the “// of all 3 lines”).

Sudo systemctl restart bind9.service

Sudo systemctl status bind9.service

Sudo nano /etc/bind/named.conf.local

UNCOMMENT the “include” line (REMOVE the “//”).

ADD a new zone, “example.com”

zone "example.com" {

type master;

file "/etc/bind/db.example.com";

};

```
include "/etc/bind/zones.rfc1918";
zone "example.com" {
        type master;
        file "/etc/bind/db.example.com";
}
```

ls /etc/bind

sudo cp /etc/bind/db.local /etc/bind/db.example.com

Sudo nano /etc/bind/db.example.com

- Change “localhost” everywhere to “example.com”
- Change the IP address 127.0.0.1 to 192.168.1.10.

These steps configure a PRIMARY server.
Before:

```
;
; BIND data file for local loopback interface
;
$TTL    604800
@       IN      SOA     localhost. root.localhost. (
                              2         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;
@       IN      NS      localhost.
@       IN      A       127.0.0.1
@       IN      AAAA    ::1
```

After:

```
;
; BIND data file for local loopback interface
;
$TTL    604800
@       IN      SOA     example.com. root.example.com. (
                              3         ; Serial
                         604800         ; Refresh
                          86400         ; Retry
                        2419200         ; Expire
                         604800 )       ; Negative Cache TTL
;
@       IN      NS      ns.example.com.
@       IN      A       192.168.1.10
@       IN      AAAA    ::1
```

Sudo nano /etc/bind/named.conf.local

Before:

```
  GNU nano 6.2
//
// Do any local configuration here
//

// Consider adding the 1918 zones here, if they are not used in your
// organization
include "/etc/bind/zones.rfc1918";
zone "example.com" {
        type master;
        file "/etc/bind/db.example.com";
};
```

After :

```
zone "example.com" {
type master;
file "/etc/bind/db.example.com";

allow-transfer {192.168.1.11; };
also-notify {192.168.1.11; };
};

zone "1.168.192.in-addr.arpa"{
type master;
file "/etc/bind/db.192";
allow-transfer {192.168.1.11; };
also-notify {192.168.1.11; };
};
```

```
  GNU nano 6.2
//
// Do any local configuration here
//

// Consider adding the 1918 zones here, if they are not used in your
// organization
include "/etc/bind/zones.rfc1918";
zone "example.com" {
        type master;
        file "/etc/bind/db.example.com";
        allow-transfer { 192.168.1.11; };
        also-notify { 192.168.1.11; };
};

zone "1.168.192.in-addr.arpa" {
        type master;
        file "/etc/bind/db.192";
        allow-transfer { 192.168.1.11; };
        also-notify { 192.168.1.11; };
};
```

sudo cat /var/log/syslog | more

To test DNS resolution:

nslookup example.com

Practical 7: DHCP server

Sudo apt install isc-dhcp-server

Sudo nano /etc/default/isc-dhcp-server

Change INTERFACESv4 to the following: “ens33” (you can see which interface is used via ip a)

ifconfig (to check ur ip address)
Sudo nano /etc/dhcp/dhcpd.conf
1) UNCOMMENT the “authoritative” line
2) Change “option domain-name-servers” to 8.8.8.8 (this is Google’s public DNS)
3) Add the following lines below (ddns-update-style none):

```
ddns-update-style none;
subnet 192.168.232.0 netmask 255.255.255.0{
range 192.168.232.150 192.168.232.200;
option routers 192.168.232.1;
option subnet-mask 255.255.255.0;
option broadcast-address 192.168.232.255;
}
```

Sudo service isc-dhcp-server restart
Sudo service isc-dhcp-server status

Use valid IP range

- Don’t include:
  - .0 (network)
  - .1 (gateway)
  - .255 (broadcast)

Example:

192.168.232.150 – 192.168.232.200

🔹 NFS Server and Client (compulsory)

SERVER:
sudo apt-get install nfs-kernel-server
sudo mkdir -p /mnt/nfsshare
sudo nano /etc/exports

Add the following lines:
/mnt/nfsshare 192.168.136.137(rw,sync,no_subtree_check)

sudo exportfs -a
sudo systemctl restart nfs-kernel-server
sudo systemctl status nfs-kernel-server

Ifconfig (copy the ip address)
sudo ufw allow from client_IP to any port nfs
(eg. sudo ufw allow from 192.168.136.137 to any port nfs)
df -h (to show final output)

CLIENT:
sudo apt-get install nfs-common
sudo mkdir -p /nfs/general
sudo mount server_IP:/mnt/nfsshare /nfs/general
(Eg: sudo mount 192.168.136.137:/mnt/nfsshare /nfs/general)

df -h

Practical 8: MySQL (Xampp)

1. Open XAMPP Server, and click “Start” for Apache and MySQL

(If MySQL doesn't start, go to “Services” and shut down (stop) MySQL80.)

2. In MySQL, click Admin (it will open in browser)

3. In the Admin section, click “Databases”

4. Create a new Database (eg: somaiya123) and save.

(Remember: Database name is to be written in the code below)

5. Add columns(name, email, password),datatypes and size. Then click Save.

6. Write the code in a notepad file:

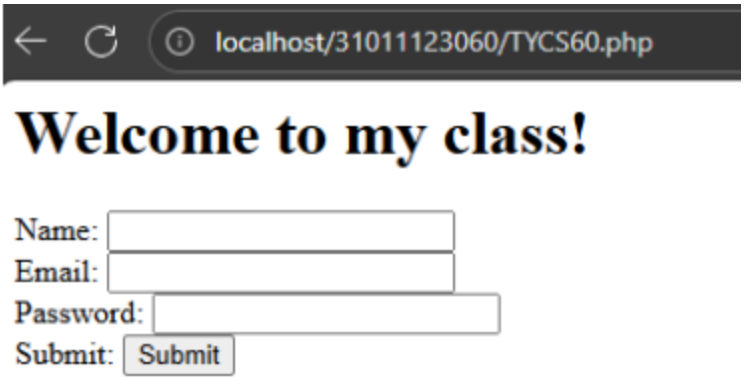
```
<html>
<head> <title> Practical 6 </title> </head>
<body>
<h1> Welcome to my class! </h1>
<form method="post">
Name: <input type="text" name="name"> <br />
Email: <input type="text" name="email"> <br />
Password: <input type="password" name="password"> <br />
<input type="submit" name="sub">
</form>
<?php
$con = mysqli_connect('localhost', 'root', '', 'somaiya123');
if(!$con){
die("Connection failed: ".mysqli_connect_error());
}
if(isset($_POST['sub'])){
$name = $_POST['name'];
```

```
$email = $_POST['email'];
$password = $_POST['password'];
$query = "INSERT INTO prac6 VALUES ('$name', '$email', '$password')";
$execute = mysqli_query($con, $query);
}
?>
</body>
</html>
```

7. Save it in the htdocs folder in C:\\xampp (ENSURE it is saved as a .php file!)

8. Then go to browser and enter localhost/31011123060/ in the URL and enter.

Note: 31011123060/ is a folder created by the user. Folder name can be anything.



Enter data and click submit.
Now go check the MYSQL (Admin) in the browser.

Practical 9: Samba

Sudo su
Sudo apt-get update
Sudo apt-get install samba

Confirm it with y!

Sudo cp /etc/samba/smb.conf /etc/samba/smb.conf.backup

Mkdir somaiya_1234
Cd somaiya_1234

Sudo adduser somaiya

Sudo smbpasswd -a somaiya

Sudo systemctl start smbd

Sudo systemctl status smbd

Sudo nano /etc/samba/smb.conf

Add the following lines at the VERY end of the file:
[FOLDER NAME]
path = /home/samba/FOLDER NAME
read only = no
browseable = yes
guest ok = yes
force user = USERNAME

Eg:

```
[somaiya_1234]
        path = /home/samba/somaiya_1234
        browseable = yes
        read only = no
        guest ok = yes
        force user=somaiya
```

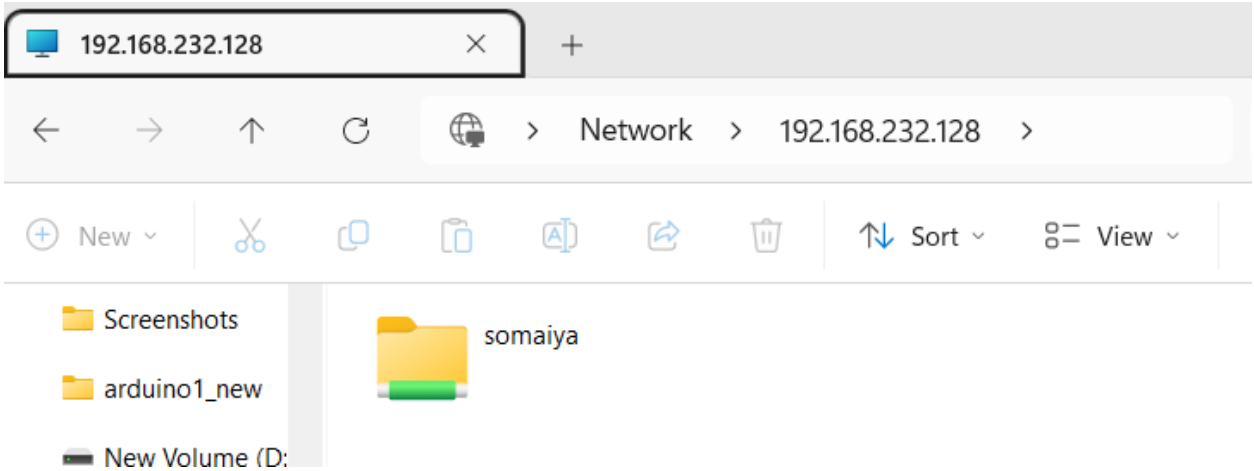
Save(ctrl + o) and exit (ctrl + x)

Sudo systemctl restart smbd

Sudo systemctl status smbd

Ifconfig

In windows file explores go to network folder and in address bar type \\ipaddress



Practical 10: Install and configure NFS server.

SERVER:

sudo apt-get install nfs-kernel-server
sudo mkdir -p /mnt/nfsshare
sudo nano /etc/exports
Ifconfig (copy the ip address)

Add the following lines:
/mnt/nfsshare 192.168.136.137(rw,sync,no_subtree_check)

sudo exportfs -a
sudo systemctl restart nfs-kernel-server
sudo systemctl status nfs-kernel-server

sudo ufw allow from client_IP to any port nfs
(eg. sudo ufw allow from 192.168.136.137 to any port nfs)

df -h (to show final output)

CLIENT:
sudo apt-get install nfs-common
sudo mkdir -p /nfs/general
sudo mount server_IP:/mnt/nfsshare /nfs/general
(Eg: sudo mount 192.168.136.137:/mnt/nfsshare /nfs/general)

df -h

AUTOMOUNT:
For automount to automatically mount instead (the question will specify this, or ma'am will ask if needed):

sudo apt-get install autofs
sudo nano /etc/auto.master

Add the line:

- /mnt/auto /etc/auto.misc --timeout=60

sudo nano /etc/auto.misc

Add the line:

- sharename -rw,sync,no_subtree_check host_IP://mnt/nfsshare

sudo systemctl restart autofs
sudo systemctl status autofs

cd /mnt/auto/sharename
df -h

Practical 11: Mounting and Parting

lsblk (displays a list of all blocks)

sudo mkfs.ext4 /dev/sda1
sudo mkdir /data
sudo mount /dev/sda1 /data

df -h (reveals that /data is Mounted on /dev/sda1)

lsblk (also shows the file system)
lsblk -f (shows label names as well)

sudo e2label /dev/sda1 data
nano filescript.sh

```
echo "enter partition u want to find: "
read PART

if [[ -b "$PART" ]]; then
    echo "Partition exists"

    lsblk -no FSTYPE,MOUNTPOINT,LABEL "$PART"
else
    echo "Partition does not exist"
fi
```

Save it using ctrl + O and exit using ctrl + X

chmod +x filescript.sh (to give execution rights)
./filescript.sh (to execute)

Enter partition:
/dev/sda1

```
siddhant@siddhant-virtual-machine:~$ ./filescript.sh
Enter the partition you want to find:
/dev/sda1
Partition /dev/sda1 exists
Enter the file system type you want to find:
File system type ext4 exists
Mount directory /data exists
data
siddhant@siddhant-virtual-machine:~$ ./filescript.sh
Enter the partition you want to find:
gdfgd
Partition gdfgd does not exist
```

EXTRA Practical: Netstat

🌐 1. PING COMMANDS

🔹 ping www.google.com

👉 Checks connectivity with a server

🔹 ping [destination]

👉 Sends packets to test if a host is reachable

🔹 ping -c [count] [destination]

👉 Sends a **fixed number of pings**
✔️ Example: -c 5 → sends 5 packets

🔹 ping -s [size] [destination]

👉 Sends packets of a **specific size**

🔹 ping -i [time] [destination]

👉 Sets **interval between pings**

🔹 ping -w [time] [destination]

👉 Stops ping after given **time duration**

🔹 ping -f [destination]

👉 Sends **flood pings** (used for testing network performance ⚠️)

📡 2. NETSTAT COMMANDS

🔹 netstat -a

👉 Shows **all active connections & listening ports**

🔹 netstat -at

👉 Shows **only TCP connections**

🔹 netstat -au

👉 Shows **only UDP connections**

🔹 netstat -l

👉 Shows **listening ports only**

🔹 netstat -lx

👉 Shows **UNIX socket ports**

🔹 netstat -s

👉 Displays **network statistics (all)**

🔹 netstat -st

👉 Shows **TCP statistics**

🔹 netstat -su

👉 Shows **UDP statistics**

🔹 netstat -pt

👉 Shows **TCP connections with process info**

🔹 netstat -pu

👉 Shows **UDP connections with process info**

🔹 netstat -c

👉 Displays output **continuously (live monitoring)**

🔹 netstat -r

👉 Shows **routing table (kernel routing info)**

🔹 netstat -ap | grep ssh

👉 Filters and shows **only SSH-related connections**

🔹 netstat -an | grep 960

👉 Finds connections related to **port 960**

🌍 3. IFCONFIG COMMANDS

🔹 ifconfig

👉 Displays **network interface details**

🔹 ifconfig -a

👉 Shows **all interfaces (even inactive ones)**

🔹 ifconfig -s

👉 Shows **short summary of interfaces**

🌐 4. DNS COMMANDS

🔹 dig [domain]

👉 Gets **DNS information (IP, records, etc.)**

🔹 nslookup

👉 Queries **DNS server for domain info**

🔹 set type=a

👉 Fetches **A record (IPv4 address)**

🔹 set type=cname

👉 Fetches **CNAME (alias record)**

🚦 5. ROUTE COMMANDS

🔹 route

👉 Displays **routing table**

🔹 route -n

👉 Shows routing table **without resolving names (faster)**

🔹 sudo route add default gw [IP]

👉 Adds a **default gateway**

🔹 sudo route add -host [IP] reject

👉 Blocks a specific host (marks as unreachable)

🔹 ip route

👉 Displays **modern routing info (replacement of route)**

🔹 sudo route del default

👉 Deletes **default gateway**